Continuous function

The function f is continuous at x_0

① $f(x_0)$ is defined

② $\lim_{x \to x_0} f(x)$ exists

③ $\lim_{x \to x_0} f(x) = f(x_0)$

δ-ε definition: $\forall \varepsilon > 0$, $\exists \delta > 0$ such that $|x - x_0| < \delta$
$\Rightarrow |f(x) - f(x_0)| < \varepsilon$

Thm: f is continuous $[a, b]$ $\Rightarrow$ f is bounded

Thm: f is continuous $[a, b]$ $\Rightarrow$ f achieves both minimum and maximum

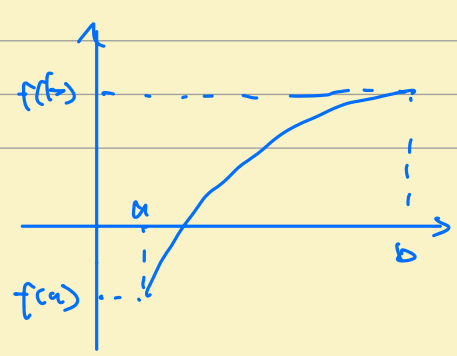
介值定理

Thm (Intermediate Value Theorem) Let $f: [a, b] \to \mathbb{R}$ be a continuous function, suppose that $f(a) \neq f(b)$. if μ is a value that $f(a) < \mu < f(b)$ ($f(b) < \mu < f(a)$), then there exists x_0 such that

$$f(x_0) = \mu \qquad x_0 \in (a, b)$$

Corollary (Existance of root) Let $f: [a, b] \to \mathbb{R}$ be a continuous function such that $f(a) < 0$ and $f(b) > 0$ (or $f(a) > 0$, $f(b) < 0$) then there exists x_0 such that

$$f(x_0) = 0 \qquad x_0 \in (a, b)$$

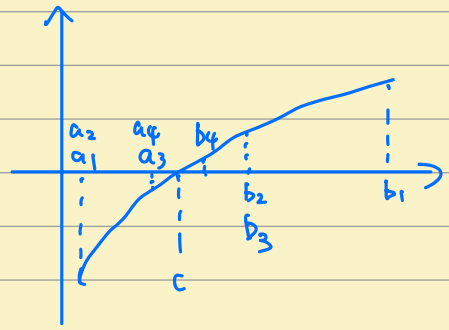


Proof: We define two sequence $\{a_n\}$ and $\{b_n\}$

① $a_1 := a$ and $b_1 := b$

② If $f\left(\frac{a_n+b_n}{2}\right) \geq 0$, let $a_{n+1} = a_n$ and $b_{n+1} = \frac{a_n+b_n}{2}$

③ If $f\left(\frac{a_n+b_n}{2}\right) < 0$, let $a_{n+1} = \frac{a_n+b_n}{2}$ and $b_{n+1} = b_n$



(i) if $a_n < b_n \Rightarrow a_{n+1} < b_{n+1}$

for all $n \in \mathbb{N}$ $a_n < b_n$

(ii) $\{a_n\}$ and $\{b_n\}$ are monotone

$a_n \leq a_{n+1}$ $b_n \geq b_{n+1}$

(iii) $\{a_n\}$ and $\{b_n\}$ are bounded

By Monotone convergence Theorem: Suppose $\{x_n\}$ is a monotone sequence, then $\{x_n\}$ is convergent if and only if $\{x_n\}$ is bounded.

Suppose $\lim_{n\to\infty} a_n = c$ $\lim_{n\to\infty} b_n = d$

Case ② $b_{n+1} - a_{n+1} = \frac{b_n+a_n}{2} - a_n = \frac{b_n - a_n}{2}$

case ③ $b_{n+1} - a_{n+1} = b_n - \frac{a_n+b_n}{2} = \frac{b_n - a_n}{2}$

$$\Rightarrow \quad b_{n+1} - a_{n+1} = \frac{1}{2}(b_n - a_n)$$
$$= \left(\frac{1}{2}\right)^2 (b_{n-1} - a_{n-1}) = \frac{1}{2^{n-1}}(b-a)$$

$$\lim_{n\to\infty} b_n - \lim_{n\to\infty} a_n = \lim_{n\to\infty}\left(\frac{1}{2^{n-1}}(b-a)\right) = 0$$

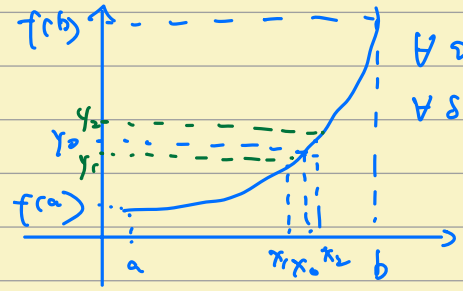
$\lim_{n\to\infty} b_n = \lim_{n\to\infty} a_n = x_0$ $f(b_n) \geq 0$, $f(a_n) \leq 0$

$$\lim_{b_n\to x_0} f(b_n) = \lim_{a_n\to x_0} f(a_n) = 0$$

Thm (Continuity of the inverse function) Let $f:[a,b]\to\mathbb{R}$ be a continuous function, if f is monotonic, then its inverse function f^{-1} is a continuous function on $[f(a), f(b)]$.

Proof: Since f is continuous, suppose f is monotonically increasing.

$y_0 \in (f(a), f(b))$, let $x_0 = f^{-1}(y_0)$ $\quad x_0 \in (a,b)$



$\forall \varepsilon>0\ \exists \delta>0 \Rightarrow |x-x_0|<\delta \Rightarrow |f(x)-f(x_0)|<\varepsilon$

$\forall \delta>0\ \exists \varepsilon>0 \Rightarrow |f^{-1}(y)-f^{-1}(y_0)|<\varepsilon \Rightarrow |x-x_0|<\delta$

Let $\delta = \min(y_2-y_0,\ y_0-y_1)$

$y\in(y_0-\delta,\ y_0+\delta)$

$\varepsilon = \min(x_2-x_0,\ x_1-x_0)$ $\quad x\in(x_0-\varepsilon,\ x_0+\varepsilon)$

$|f^{-1}(y)-f^{-1}(y_0)| = |x-x_0|<\varepsilon$ $\qquad f^{-1}$ is continuous at y_0.

一致连续性

Uniform continuity.

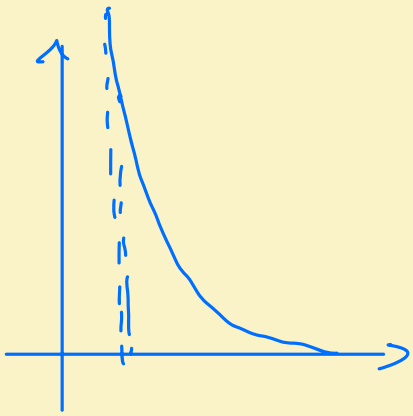
Defn (Uniform continuity) Let $f: S\to\mathbb{R}$ be a function. Suppose for any $\varepsilon>0$, there exists $\delta>0$, such that $x_1, x_2\in S$, and $|x_1-x_2|<\delta$, then

$$|f(x_1)-f(x_2)|<\varepsilon.$$

Then we say f is uniformly continuous.

Defn (Cauchy Convergence criterion). A sequence $\{x_n\}$ is convergent if and only if for any $\varepsilon>0$, there exists $N\in\mathbb{N}$, such that $n_1, n_2>N$, then

$$|x_{n_1}-x_{n_2}|<\varepsilon$$



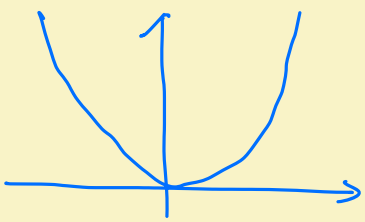
E.g. $f(x) = \frac{1}{x} : (0,1) \to \mathbb{R}$. $f(x)$ is continuous but not uniformly continuous.

Proof: Given $\varepsilon > 0$, then for $|f(x_1) - f(x_2)| < \varepsilon$ to hold, we have

$$\varepsilon > \left| \frac{1}{x_1} - \frac{1}{x_2} \right| = \left| \frac{x_2 - x_1}{x_1 x_2} \right| = \frac{|x_2 - x_1|}{x_1 x_2}$$

$$\Rightarrow |x_2 - x_1| < \varepsilon \cdot x_1 x_2$$

We must have $\delta \leq \varepsilon \cdot x_1 x_2$ for all $x_1, x_2 \in (0,1)$, then $\delta \leq 0$ by definition of uniform continuity $\delta > 0$. Contradiction.



E.g. $f(x) = x^2 : \mathbb{R} \to \mathbb{R}$ is not uniformly continuous.

Proof: $a_n = n + \frac{1}{n}$, $b_n = n$

$$\lim_{n\to\infty} (a_n - b_n) = \lim_{n\to\infty} \left(n + \frac{1}{n} - n\right) = \lim_{n\to\infty} \frac{1}{n} = 0$$

$x_1 = a_n$, $x_2 = b_n$, we have $|x_1 - x_2| < \delta$ for $n > N$

but $$|f(x_1) - f(x_2)| = |f(a_n) - f(b_n)|$$

$$= \left| \left(n + \frac{1}{n}\right)^2 - n^2 \right| = \left| 2 + \frac{1}{n^2} \right| \geq 2$$

Then f is not uniformly continuous on $\mathbb{R}$.

$|x_1 + x_2| \leq 2$

$f(x) = x^2 : [a,b] \to \mathbb{R}$ is uniformly continuous.

Proof: Let $[a,b] = [0,1]$ take $0 \leq x_1, x_2 \leq 1$

$$|f(x_1) - f(x_2)| = |x_1^2 - x_2^2| = |x_1 + x_2| |x_1 - x_2| \leq 2|x_1 - x_2|$$

Therefore, given $\varepsilon > 0$, let $\delta = \frac{\varepsilon}{2}$. If $|x_1 - x_2| < \delta$, $\Rightarrow |f(x_1) - f(x_2)| < \varepsilon$.

$f(x) = \frac{1}{x}$ is uniformly continuous on $[1, +\infty)$

E.g. Let $f(x): S \to \mathbb{R}$ be a function, then $f(x)$ is uniformly continuous if and only if. for any sequences $\{x_n^1\}, \{x_n^2\} \in S$,

$$\lim_{n\to\infty}(x_n^1 - x_n^2) = 0 \quad \Rightarrow \quad \lim_{n\to\infty}\left(f(x_n^1) - f(x_n^2)\right) = 0$$

Proof: ($\Rightarrow$) $\forall \varepsilon > 0, \exists \delta > 0 \ \forall x_n^1, x_n^2 \in S, |x_n^1 - x_n^2| < \varepsilon \Rightarrow |f(x_n^1) - f(x_n^2)| < \delta$

if $\lim_{n\to\infty}(x_n^1 - x_n^2) = 0 \Rightarrow \lim_{n\to\infty}|f(x_n^1) - f(x_n^2)| = 0$.

($\Leftarrow$) Suppose that $f(x)$ is not uniformly continuous.

$\exists \varepsilon > 0, \forall \delta > 0, \exists x^1, x^2 \in S$ such that $|x^1 - x^2| < \delta$, but $|f(x^1) - f(x^2)| \geq \varepsilon$.

Let $\delta_1 = 1 \quad \exists \ |x_1^1 - x_1^2| < \delta_1, \ |f(x_1^1) - f(x_1^2)| \geq \varepsilon$

Let $\delta_2 = \frac{1}{2} \quad \exists \ |x_2^1 - x_2^2| < \delta_2, \ |f(x_2^1) - f(x_2^2)| \geq \varepsilon$

$\vdots$

Let $\delta_n = \frac{1}{n} \quad \exists \ |x_n^1 - x_n^2| < \delta_n, \ |f(x_n^1) - f(x_n^2)| \geq \varepsilon$

$\lim_{n\to\infty}|x_n^1 - x_n^2| = 0$ but $\lim_{n\to\infty} f(x_n^1) - f(x_n^2) \neq 0$

5.X.

① Expand the following function and make them be continuous on $\mathbb{R}$.

(1). $f(x) = \dfrac{x^3 - 8}{x - 2}$ $\qquad f(x) = \begin{cases} \dfrac{x^3 - 8}{x - 2} & x \neq 2 \\ 12 & x = 2 \end{cases}$

(2) $f(x) = \dfrac{1 - \cos x}{x^2}$ $\qquad f(x) = \begin{cases} \dfrac{1 - \cos x}{x^2} & x \neq 0 \\ \dfrac{1}{2} & x = 0 \end{cases}$

(3) $f(x) = x \cos\frac{1}{x}$. $\qquad f(x) = \begin{cases} x\cos\frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$

② Prove that if f, g are uniformly continuous on $[a, b]$, then $f-g$ is uniformly continuous on $[a, b]$

$\forall \varepsilon > 0$, $\exists \delta_1 > 0$ such that $|x_1 - x_2| < \delta_1 \Rightarrow |f(x_1) - f(x_2)| < \frac{\varepsilon}{2}$

$\forall \varepsilon > 0$, $\exists \delta_2 > 0$ such that $|x_1 - x_2| < \delta_2 \Rightarrow |g(x_1) - g(x_2)| < \frac{\varepsilon}{2}$

Let $\delta = \min(\delta_1, \delta_2)$ $\qquad |a-b| \le |a| + |b|$

$$|(f-g)(x_1) - (f-g)(x_2)| = |(f(x_1) - f(x_2)) - (g(x_1) - g(x_2))|$$

$$\le |f(x_1) - f(x_2)| + |g(x_1) - g(x_2)| < \varepsilon$$

$\forall \varepsilon > 0$, $\exists \delta = \min(\delta_1, \delta_2)$ such that $|x_1 - x_2| < \delta$

$$\Rightarrow |(f-g)(x_1) - (f-g)(x_2)| < \varepsilon$$

③ Suppose that function f satisfies the Lipschitz condiction:

$$|f(x_1) - f(x_2)| \le L|x_1 - x_2|$$

$L > 0$, $x_1, x_2 \in [a, b]$, then f is uniformly continuous on $[a, b]$,

$\forall \varepsilon > 0$, $\exists \delta > 0$ such that $|x_1 - x_2| < \delta \Rightarrow |f(x_1) - f(x_2)| < \varepsilon$.

$$|f(x_1) - f(x_2)| \le L|x_1 - x_2| < L\delta = \varepsilon \qquad \varepsilon = L\delta$$

Let $\delta = \frac{\varepsilon}{L}$, $|x_1 - x_2| < \delta$

$$\Rightarrow |f(x_1) - f(x_2)| < L|x_1 - x_2| < L \cdot \delta = L \cdot \frac{\varepsilon}{L} = \varepsilon$$

Continuity of Elementary functions.

基本初等函数

Basic Elementary functions

(1) $f(x) = C$ $\qquad$ (2) $f(x) = x$ $\qquad$ (3) $f(x) = x^a$ $\quad a \in \mathbb{R}$

(4) $f(x) = \log_a x$ $\qquad$ (5) $f(x) = \sin x$

$f(x) = \cos x$

$f(x) = \tan x$

$f(x) = \cot x$

$f(x) = \arcsin x$

$f(x) = \arccos x$

$f(x) = \arctan x$

$f(x) = \operatorname{arccot} x$

Thm: Every basic elementary function is continuous on its domain.

Thm: Every elementary function that is obtained by algebra from basic elementary functions is continuous on its domain.

E.g. $\lim_{x\to 0} \frac{\ln(1+x)}{x} = \lim_{x\to 0} \ln(1+x)^{\frac{1}{x}} = \ln\left(\lim_{x\to 0}(1+x)^{\frac{1}{x}}\right) = \ln e = 1$

$\lim_{x\to x_0} f(x) = f(\lim_{x\to x_0} x_0)$ $\quad \lim_{x\to x_0} f(g(x)) = f(\lim_{x\to x_0} g(x))$

Thm (Theorem of uniform continuity) If function f is continuous on $[a,b]$, then f is a uniformly continuous function on $[a,b]$.

Proof: Prove by contradiction. Suppose f is not uniformly continuous but f is continuous on $[a,b]$. By definition of uniform continuity:

$\exists\ \varepsilon_0 > 0$ such that $\forall\ \delta > 0$, $\exists\ x, y \in S$, and $|x-y| < \delta \Rightarrow |f(x) - f(y)| \geq \varepsilon_0$

Since δ is arbitrary,

$\exists\ \varepsilon_0 > 0$ such that $\exists\ x_1, y_1 \in S$, and $|x_1 - y_1| < \frac{1}{1} \Rightarrow |f(x_1) - f(y_1)| \geq \varepsilon_0$.

$\exists\ \varepsilon_0 > 0$ such that $\exists\ x_2, y_2 \in S$, and $|x_2 - y_2| < \frac{1}{2} \Rightarrow |f(x_2) - f(y_2)| \geq \varepsilon_0$.

$\vdots$

$\exists\ \varepsilon_0 > 0$ such that $\exists\ x_n, y_n \in S$, and $|x_n - y_n| < \frac{1}{n} \Rightarrow |f(x_n) - f(y_n)| \geq \varepsilon_0$.

By above, we find sequence $\{x_n\}, \{y_n\} \in [a,b]$. By Bolzano-Weierstrass Theorem, there exists a convergent subsequence $\{x_{n_k}\}$ such that $\lim_{k\to\infty} x_{n_k} = c$ then

$$|y_{n_k} - c| = |y_{n_k} - x_{n_k} + x_{n_k} - c| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - c| < \frac{1}{n_k} + |x_{n_k} - c|$$

As both $\frac{1}{n_k}$ and $|x_{n_k} - c|$ converge to zero, $\{y_{n_k}\}$ converges to c. And

$$|f(x_{n_k}) - f(c)| = |f(x_{n_k}) - f(y_{n_k}) + f(y_{n_k}) - f(c)|$$
$$\geq |f(x_{n_k}) - f(y_{n_k})| - |f(y_{n_k}) - f(c)|$$
$$\geq \varepsilon_0 - |f(y_{n_k}) - f(c)|$$

$$\Rightarrow \quad |f(x_{n_k}) - f(c)| + |f(y_{n_k}) - f(c)| \geq \varepsilon_0$$

At least one of the sequences $\{f(x_{n_k})\}$ or $\{f(y_{n_k})\}$ does not converge. Thus f is not continuous. Contradiction.

E. x:

$$\lim_{x\to+\infty}\left(\sqrt{x+\sqrt{x+\sqrt{x}}} - \sqrt{x}\right)$$

$$= \lim_{x\to+\infty}\frac{\left(\sqrt{x+\sqrt{x+\sqrt{x}}} - \sqrt{x}\right)\left(\sqrt{x+\sqrt{x+\sqrt{x}}} + \sqrt{x}\right)}{\sqrt{x+\sqrt{x+\sqrt{x}}} + \sqrt{x}}$$

$$= \lim_{x\to+\infty}\frac{\sqrt{x+\sqrt{x}}}{\sqrt{x+\sqrt{x+\sqrt{x}}} + \sqrt{x}} = \lim_{x\to\infty}\frac{\sqrt{1+\frac{\sqrt{x}}{x}}}{\sqrt{1+\frac{\sqrt{x+\sqrt{x}}}{x}}+1} = \frac{1}{2}$$

$$\lim_{x\to0}(1+\sin x)^{\cot x} \qquad \lim_{x\to0}(1+x)^{\frac{1}{x}} = e$$

$$= \lim_{x\to0}(1+\sin x)^{\frac{\cos x}{\sin x}} \qquad u = \sin x \qquad x\to0 \quad u\to0$$

$$\cos x = \cos(\arcsin u) = \sqrt{1-u^2}$$

$$= \lim_{u\to0}(1+u)^{\frac{\sqrt{1-u^2}}{u}} \qquad x = \arcsin u$$

$$= \lim_{u\to0}\Big[\underbrace{(1+u)^{\frac{1}{u}}}_{e}\Big]^{\sqrt{1-u^2}} = \lim_{u\to0}\left[(1+u)^{\frac{1}{u}}\right]^{\lim_{u\to0}\sqrt{1-u^2}} = e^1$$